

Splunk TTP Detection — Cheat Sheet

HTB Academy | Detecting Attacker Behavior With Splunk Based On TTPs

PART 1 — Known TTP-Based Detections

Searches built on known attacker behaviors and techniques.

1. Reconnaissance via Native Windows Binaries

Attackers use built-in Windows tools like net.exe, whoami.exe, and ipconfig.exe to profile the environment, enumerate users, check network config, and identify lateral movement opportunities — all without dropping any external tools.

```
index="main" sourcetype="WinEventLog:Sysmon" EventCode=1
Image=*\\ipconfig.exe OR Image=*\\net.exe OR Image=*\\whoami.exe
OR Image=*\\netstat.exe OR Image=*\\nbtstat.exe
OR Image=*\\hostname.exe OR Image=*\\tasklist.exe
| stats count by Image, CommandLine | sort - count
```

2. Payload Hosting on githubusercontent.com

Attackers use GitHub raw content as a payload delivery platform because it is commonly whitelisted by company proxies. DNS queries for github domains from non-browser processes are a strong signal of malicious downloads.

```
index="main" sourcetype="WinEventLog:Sysmon" EventCode=22
QueryName=*github*
| stats count by Image, QueryName
```

3. PsExec Detection — Registry Modification (Event ID 13)

PsExec installs a service on the remote host by modifying the ImagePath registry key under HKLM\System\CurrentControlSet\Services. Seeing services.exe write to this key for an unexpected binary is a strong PsExec indicator.

```
index="main" sourcetype="WinEventLog:Sysmon" EventCode=13
Image="C:\\Windows\\system32\\services.exe"
TargetObject="HKLM\\System\\CurrentControlSet\\Services\\*\\ImagePath"
| rex field=Details "(?[^\\]+)$"
| eval reg_file_name=lower(reg_file_name),
file_name=if(isnull(file_name),reg_file_name,lower(file_name))
| stats values(Image) AS Image, values(Details) AS RegistryDetails,
values(_time) AS EventTimes, count by file_name, ComputerName
```

4. PsExec Detection — File Drop via Admin\$ (Event ID 11)

PsExec copies PSEXESVC.exe to the remote machine via the hidden Admin\$ share. The System process creating files outside normal directories is a reliable indicator of this lateral movement technique.

```
index="main" sourcetype="WinEventLog:Sysmon" EventCode=11
Image=System
| stats count by TargetFilename
```

5. PsExec Detection — Named Pipe Communication (Event ID 18)

PsExec establishes communication between the attacker and the remote service using named pipes. Pipe connections from the System process to PsExec-style pipe names indicate active remote execution.

```
index="main" sourcetype="WinEventLog:Sysmon" EventCode=18
Image=System
| stats count by PipeName
```

6. Archive Files Used for Exfiltration or Tool Transfer

Attackers compress tools or stolen data into zip, rar, or 7z archives before transferring them. Unexpected archive creation — especially in user temp folders or desktops — warrants investigation.

```
index="main" EventCode=11
(TargetFilename="*.zip" OR TargetFilename="*.rar" OR TargetFilename="*.7z")
| stats count by ComputerName, User, TargetFilename | sort - count
```

7. PowerShell Downloading Payloads

PowerShell is a top tool for downloading additional malware stages. Files written to disk by powershell.exe — especially executables or scripts — are a reliable indicator of a download-and-execute attack pattern.

```
index="main" sourcetype="WinEventLog:Sysmon" EventCode=11
Image="*powershell.exe*"
| stats count by Image, TargetFilename | sort + count
```

8. MS Edge Downloading Files from Internet

Zone.Identifier is an Alternate Data Stream Windows attaches to files downloaded from the internet. Browser downloads of executables, scripts, or Office documents are high-risk and should be reviewed.

```
index="main" sourcetype="WinEventLog:Sysmon" EventCode=11
Image="*msedge.exe" TargetFilename="*Zone.Identifier"
| stats count by TargetFilename | sort + count
```

9. Execution from Downloads Folder

Legitimate software almost never executes directly from a user Downloads folder. Execution here strongly suggests a file was dropped via phishing, browser exploit, or manual attacker action.

```
index="main" EventCode=1
| regex Image="C:\\\\Users\\\\.*\\\\Downloads\\\\.*"
| stats count by Image
```

10. Executables or DLLs Created Outside Windows Directory

Malware frequently drops EXE or DLL files in user-writable directories like AppData, Temp, or Public to avoid detection. Files created outside the Windows directory tree by unexpected processes are suspicious.

```
index="main" EventCode=11
(TargetFilename="*.exe" OR TargetFilename="*.dll")
TargetFilename!="*\\windows\\"
| stats count by User, TargetFilename | sort + count
```

11. Misspelled Legitimate Binaries (Masquerading)

Attackers name their malicious binaries to resemble legitimate ones — e.g. psexesvc.exe vs PSEXESVC.exe. This catches variations of psexec in command lines that are NOT the known legitimate PsExec binaries.

```
index="main" sourcetype="WinEventLog:Sysmon" EventCode=1
(CommandLine="*psexec*.exe"
NOT (CommandLine="*PSEXESVC.exe" OR CommandLine="*PsExec64.exe"))
OR (ParentCommandLine="*psexec*.exe"
NOT (ParentCommandLine="*PSEXESVC.exe" OR ParentCommandLine="*PsExec64.exe"))
OR (Image="*psexec*.exe"
NOT (Image="*PSEXESVC.exe" OR Image="*PsExec64.exe"))
| table Image, CommandLine, ParentImage, ParentCommandLine
```

12. Non-Standard Port Communications

Attackers use non-standard ports to blend C2 traffic with normal activity or bypass firewall rules. This query excludes the most common legitimate ports (HTTP, HTTPS, SSH, FTP) to surface unusual outbound connections.

```
index="main" EventCode=3
NOT (DestinationPort=80 OR DestinationPort=443
OR DestinationPort=22 OR DestinationPort=21)
| stats count by SourceIp, DestinationIp, DestinationPort | sort - count
```

PART 2 — Analytics & Anomaly-Based Detections

Searches that use statistical analysis to surface abnormal behavior — not just known signatures.

1. Abnormally Long Command Lines

Attackers often use very long commands to obfuscate their activity, pass encoded payloads, or chain multiple actions. Sorting by command length surfaces the longest — and potentially most suspicious — commands. Filtering out `msiexec.exe` and `explorer.exe` removes the most common benign sources of long commands.

```
index="main" sourcetype="WinEventLog:Sysmon" Image=*cmd.exe
ParentImage!="*msiexec.exe" ParentImage!="*explorer.exe"
| eval len=len(CommandLine)
| table User, len, CommandLine | sort - len
```

2. Abnormal cmd.exe Execution Volume (Statistical Outlier Detection)

This query groups `cmd.exe` executions into 1-hour buckets, calculates the average and standard deviation across the environment, then flags any hour/user/command combination that exceeds 1.5 standard deviations above the average. This surfaces bursts of `cmd.exe` activity that are statistically unusual — a common sign of automated attacker tooling or scripted lateral movement.

```
index="main" EventCode=1 (CommandLine="*cmd.exe")
| bucket _time span=1h
| stats count as cmdCount by _time User CommandLine
| eventstats avg(cmdCount) as avg stdev(cmdCount) as stdev
| eval isOutlier=if(cmdCount > avg+1.5*stdev, 1, 0)
| search isOutlier=1
```

3. Processes Loading an Unusually High Number of DLLs

Malware — especially loaders and injectors — often loads many DLLs in rapid succession during execution. This query counts distinct DLLs loaded per process per hour, then surfaces processes loading more than 3 unique DLLs in that window. Known benign paths (`System32`, `Program Files`, `ProgramData`, `AppData`) are excluded to reduce noise.

```
index="main" EventCode=7
NOT (Image="C:\\Windows\\System32*")
NOT (Image="C:\\Program Files (x86)*")
NOT (Image="C:\\Program Files*")
NOT (Image="C:\\ProgramData*")
NOT (Image="C:\\Users\\waldo\\AppData*")
| bucket _time span=1h
| stats dc(ImageLoaded) as unique_dlls_loaded by _time, Image
| where unique_dlls_loaded > 3
| stats count by Image, unique_dlls_loaded
| sort - unique_dlls_loaded
```

4. Same Process Created More Than Once on the Same Machine

The transaction command groups all process creation events sharing the same `Image` and `ComputerName`. If more than one unique `ProcessGuid` exists within that group, the same executable was launched multiple times on the same host. This can indicate beaconing, repeated execution of a dropper, or an attacker re-running tools. Drilling into specific pairs (like `rundll32` → `svchost`) exposes the actual command lines used.

```
index="main" sourcetype="WinEventLog:Sysmon" EventCode=1
| transaction ComputerName, Image
| where mvcount(ProcessGuid) > 1
| stats count by Image, ParentImage
```

4b. Drill Down — rundll32.exe spawned by svchost.exe

After identifying suspicious image-parent pairs, drill down to see the exact command lines. `rundll32.exe` spawned repeatedly by `svchost.exe` with unusual arguments is a classic living-off-the-land technique used for code execution and persistence.

```
index="main" sourcetype="WinEventLog:Sysmon" EventCode=1
| transaction ComputerName, Image
| where mvcount(ProcessGuid) > 1
| search Image="C:\\Windows\\System32\\rundll32.exe"
ParentImage="C:\\Windows\\System32\\svchost.exe"
```

```
| table CommandLine, ParentCommandLine
```

Key Sysmon Event IDs Reference

Event ID	Name	Detection Use
1	Process Creation	Parent-child anomalies, execution from unusual paths, command line analysis
3	Network Connection	Non-standard ports, suspicious outbound connections, C2 detection
7	Image Loaded	DLL load monitoring, DLL hijacking detection
10	ProcessAccess	LSASS handle access, credential dumping, shellcode in CallTrace
11	File Create	Malware drops, archive creation, browser downloads, Zone.Identifier
13	Registry Value Set	Service ImagePath modification (PsExec), persistence via Run keys
18	Pipe Connected	PsExec named pipe communication, lateral movement
22	DNS Query	Payload hosting domains, C2 beacon DNS resolution
4662	AD Object Access	DCSync attack detection (requires DC audit policy enabled)

SPL Command Reference

Command	What it does
stats count by X	Count events grouped by field X — the most common aggregation command
stats dc(X) as Y	Count distinct values of field X, store as Y — used for unique DLL counts
stats avg(X), stdev(X)	Calculate average and standard deviation of field X across results
eventstats avg(X) as Y	Like stats but adds the result as a new field without collapsing events
eval len=len(X)	Calculate the character length of field X — used to find long commands
eval isOutlier=if(...)	Create a new field with a conditional value — 1 if condition is true, 0 if not
bucket _time span=1h	Group events into time buckets of 1 hour — used for time-based analysis
transaction X, Y	Group related events sharing the same values of X and Y into one record
mvcount(X)	Count the number of values in a multivalue field — used after transaction
rex field=X "pattern"	Extract a value from field X using a regex pattern — stores in named group
regex Image="pattern"	Filter events where field matches a regex — used for path pattern matching
table X, Y, Z	Display only the specified fields as a clean table in results
sort - X	Sort results by field X in descending order (- = descending, + = ascending)
search X=1	Filter results after a transformation — like WHERE in SQL
where X > Y	Filter results using an expression — more flexible than search
NOT (X=A OR X=B)	Exclude events where field X matches any of the listed values